

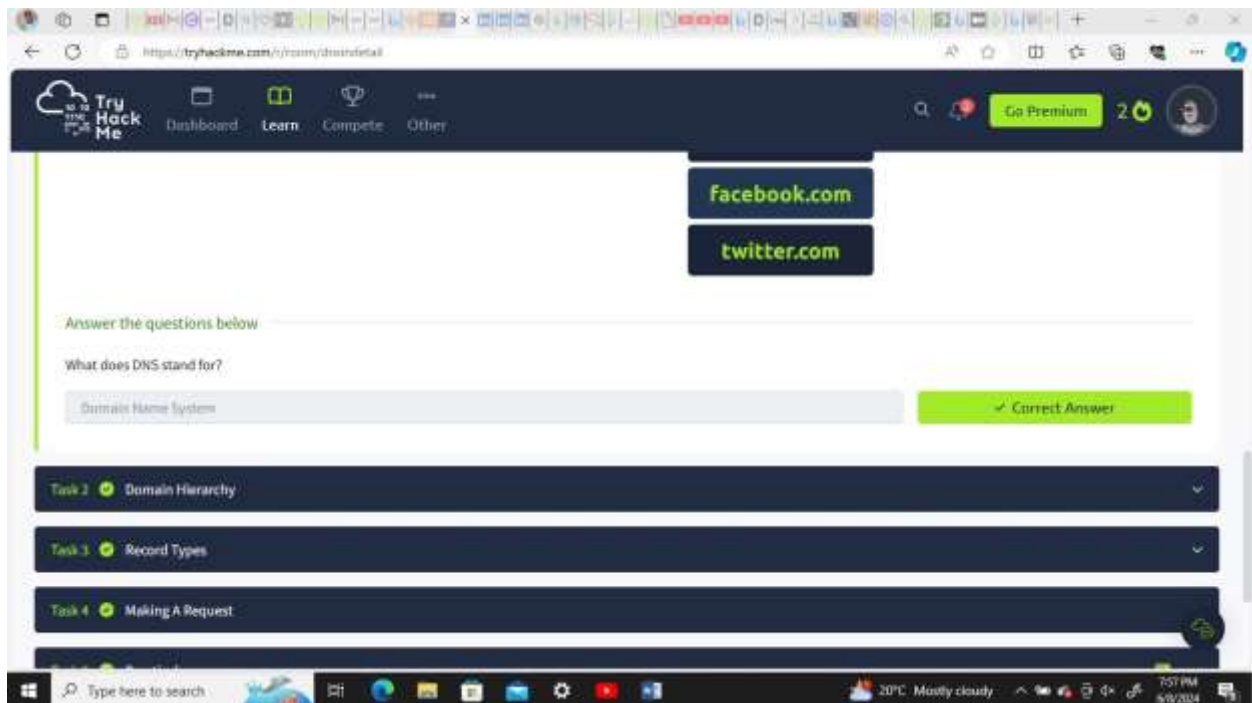
Wireshark to Investigate Network Traffic

Report by Ezekiel Waweru CNS06-24049

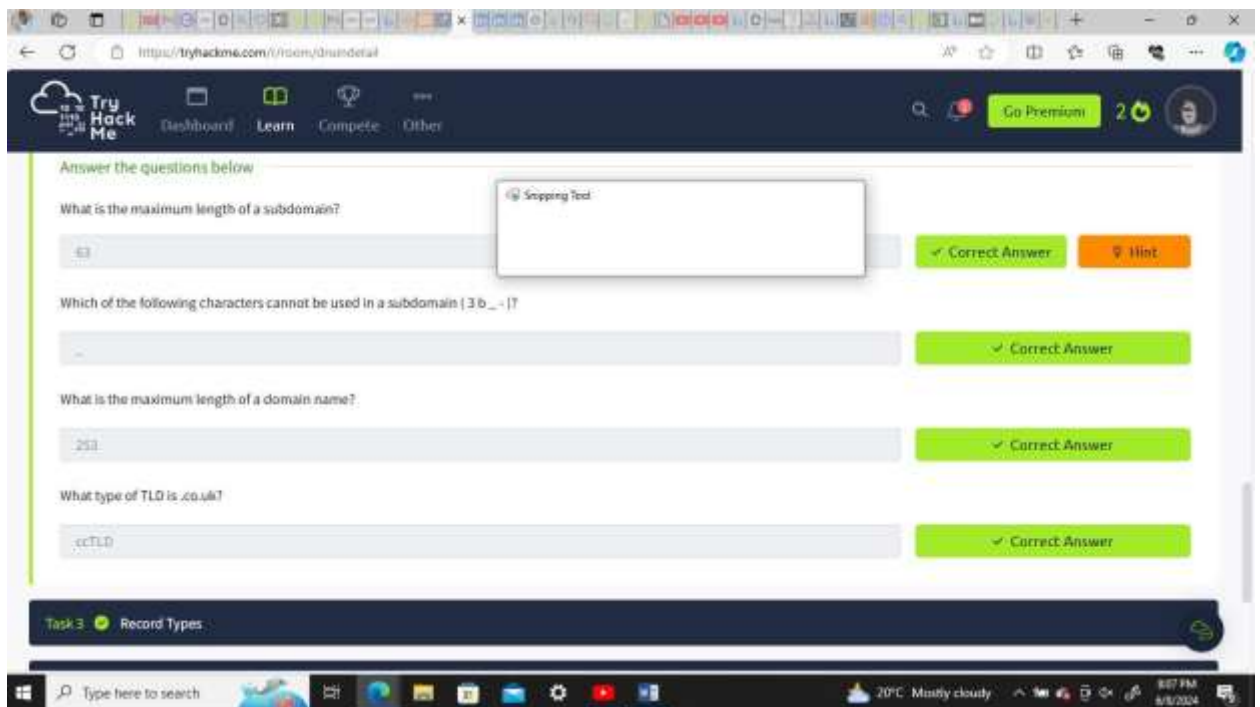
Introduction

This was an informative assignment on Domain Naming System (DNS). I learnt it as a module on TryHackMe platform. It is essential for those seeking a career in Networking or Cybersecurity to be familiar with the workings of DNS. Starting off, I thought it was a complex networking concept but the module was well articulated and I got to gain foundational insights in one of the foundational concepts of the internet.

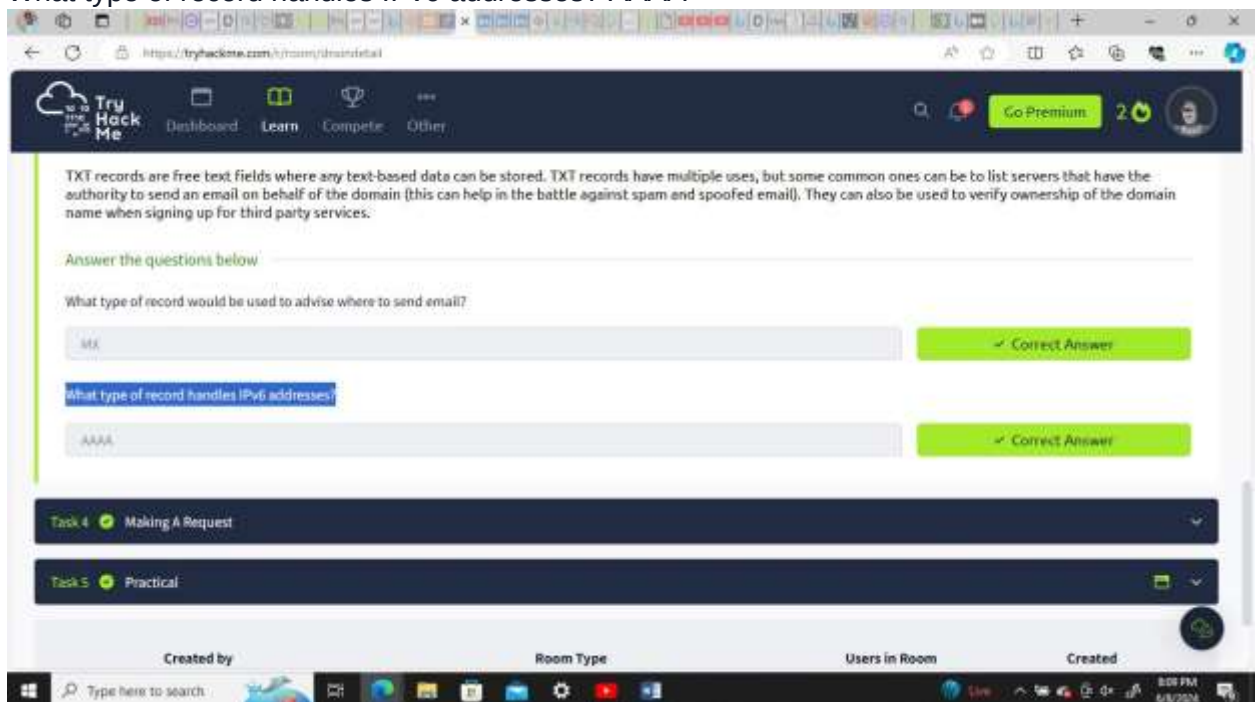
- a. What does DNS stand for? Domain name system



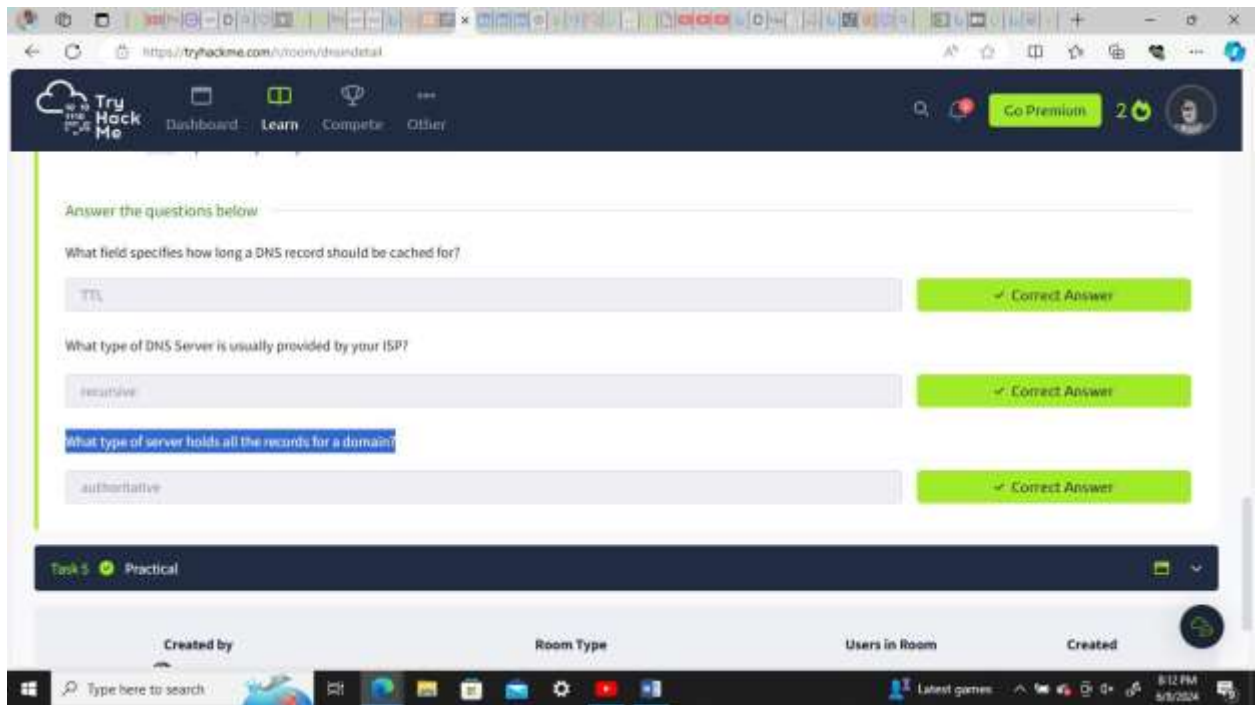
- b. What is the maximum length of a subdomain? 63
c. Which of the following characters cannot be used in a subdomain (3 b _ -)? _
d. What is the maximum length of a domain name? 253
e. What type of TLD is .co.uk? ccTLD



- f. What type of record would be used to advise where to send email? MX
- g. What type of record handles IPv6 addresses? AAAA



- h. What field specifies how long a DNS record should be cached for? TTL
- i. What type of DNS Server is usually provided by your ISP? recursive
- j. What type of server holds all the records for a domain? Authoritative



- k. What is the CNAME of shop.website.thm? shops.myshopify.com
- l. What is the value of the TXT record of website.thm? THM{7012BBA60997F35A9516C2E16D2944FF}
- m. What is the numerical priority value for the MX record? 30
- n. What is the IP address for the A record of www.website.thm? 10.10.10.10

The image displays two screenshots of a web browser window, likely a virtual machine, showing a DNS quiz and terminal output.

Top Screenshot:

- The browser address bar shows `https://tryhackme.com/r/room/dnsindetail`.
- The page title is "Room completed (100%)".
- The main content area contains a quiz titled "Answer the questions below".
- The quiz questions and answers are:
 - Question: "What is the CNAME of shop.website.thm?"
Answer: `shops.myshopify.com` (Correct Answer)
 - Question: "What is the value of the TXT record of website.thm?"
Answer: `THM{7012BBA60997F35A9516C2E1}` (Correct Answer)
 - Question: "What is the numerical priority value for the MX record?"
Answer: `30` (Correct Answer)
 - Question: "What is the IP address for the A record of www.website.thm?"
Answer: `10.10.10.10` (Correct Answer)
- A "View Site" button is visible next to the first question.
- On the right side, there is a terminal window showing DNS lookup results for `website.thm`. The terminal output includes:
 - `Address: 127.0.0.53#53`
 - `Non-authoritative answer:`
 - `website.thm text = "THM{7012BBA60997F35A9516C2E1602944FF}"`
 - `user@thm:~$ nslookup --type=MX website.thm`
 - `Server: 127.0.0.53`
 - `Address: 127.0.0.53#53`
 - `Non-authoritative answer:`
 - `website.thm mail exchanger = 30 alt4.aspmx.l.google.com`
 - `user@thm:~$ nslookup --type=A website.thm`
 - `Server: 127.0.0.53`
 - `Address: 127.0.0.53#53`
 - `Non-authoritative answer:`
 - `Name: website.thm`
 - `Address: 10.10.10.10`
 - `user@thm:~$ nslookup website.thm`

Bottom Screenshot:

- The browser address bar shows `https://tryhackme.com/r/room/dnsindetail`.
- The page title is "Room completed (100%)".
- The main content area contains a quiz titled "Answer the questions below".
- The quiz questions and answers are:
 - Question: "What is the CNAME of shop.website.thm?"
Answer: `shops.myshopify.com` (Correct Answer)
 - Question: "What is the value of the TXT record of website.thm?"
Answer: `THM{7012BBA60997F35A9516C2E1}` (Correct Answer)
 - Question: "What is the numerical priority value for the MX record?"
Answer: `30` (Correct Answer)
 - Question: "What is the IP address for the A record of www.website.thm?"
Answer: `10.10.10.10` (Correct Answer)
- A "View Site" button is visible next to the first question.
- On the right side, there is a terminal window showing DNS lookup results for `shop.website.thm`. The terminal output includes:
 - `user@thm:~$ nslookup --type=CNAME shop.website.thm`
 - `Server: 127.0.0.53`
 - `Address: 127.0.0.53#53`
 - `Non-authoritative answer:`
 - `shop.website.thm canonical name = shops.myshopify.com`
 - `user@thm:~$ nslookup --type=TXT website.thm`
 - `Server: 127.0.0.53`
 - `Address: 127.0.0.53#53`
 - `Non-authoritative answer:`
 - `website.thm text = "THM{7012BBA60997F35A9516C2E1602944FF}"`
 - `user@thm:~$ nslookup website.thm`

o. Show evidence of completion



Dashboard

Complete rooms and upskill in security, all from your browser.

Learning Path Recent Rooms

Rooms you've worked on recently



DNS in detail

Learn how DNS works and how it helps you access internet services.

Skills Matrix

As you learn, your skills matrix will grow **beta**

